

COVENANT

Agreements that read the world and pay themselves.

Somnia Agentathon · covenant-beta.vercel.app · github.com/ajanaku1/covenant

THE PROBLEM

Real deals are written in English. Contracts can't read.

- "Pay the freelancer when the site is live" has to be translated into brittle Solidity before a chain can enforce it.
- Then it needs an oracle to describe reality and a keeper or cron job to poke it forever.
- Every one of those is a middleman in a deal that was supposed to be trustless.

THE IDEA

Keep the clause in English. Make the judge a validator panel.

"Pay the freelancer 200 USDC when the site at <url> is live and mentions Somnia. Check daily for 7 days. Refund me if it isn't met by the deadline."

You fund it once. That is the last transaction anyone sends. The contract does the rest itself.

HOW IT WORKS

One loop: Wake → Perceive → Judge → Act

WAKE

The contract schedules its own wake-ups with Somnia's reactivity layer. The chain calls it at check time.

PERCEIVE

It sends the real evidence (the live website) to Somnia's LLM Parse Website agent.

JUDGE

A validator subcommittee each scores the English clause 0–100. The contract takes the median.

ACT

Score $\geq$ threshold: escrow released. Past deadline unmet: funder refunded. Then re-arm.

No single judge can swing a payout. Every ruling links to a public per-validator receipt.

DEEP SOMNIA INTEGRATION

Two primitives, used together

Reactivity	<code>scheduleSubscriptionAtTimestamp</code> (precompile 0x0100) — the contract is its own keeper. One singleton owns every schedule above a single 32 STT balance floor, never 32 per agreement.
Agent platform	<code>createAdvancedRequest</code> to the LLM Parse Website agent over a threshold-consensus subcommittee; <code>handleResponse</code> aggregates the median and settles.
Escrow safety	<code>reservedEscrow</code> walls agreement funds off from gas and agent fees. Callbacks gate on <code>msg.sender</code> and handle every <code>ResponseStatus</code> .
Self-healing	A thin buffer defers a re-arm instead of reverting a payout; permissionless poke and <code>reclaimExpired</code> recover anything the chain drops.

PROVEN ON TESTNET, NOT PROMISED

The receipts are public

- 3-validator consensus judgment resolved on-chain, median-aggregated — receipt `agents.testnet.somnia.network/receipts/5358074`
- The chain autonomously woke the contract at the exact scheduled second, in a transaction nobody sent — `0x2c4e0e06...`
- Covenant singleton live at `0x152432d1B863C0A0645D86452a23F9C16077C28A` (chain 50312)
- 32 Foundry tests: escrow math, callback discipline, refunds, deferred re-arms, recovery paths.

0

transactions after funding

1

singleton, every agreement

3

validators per judgment

WHERE THIS GOES

The freelance clause is the showcase. The engine is general.

- KPI-gated vesting: "unlock the grant when the dashboard shows 10k users."
- Vendor SLAs: "refund 5% for every day the status page shows an outage."
- Real-world settlement of any deal a person can phrase in a sentence and point at evidence.

Anything an LLM panel can read becomes an enforceable condition.

**Write a deal in a sentence.
Walk away.**

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Built for the Somnia Agentathon